

Busy Snail Filament Changer (BSFC) BOM V.2026-02-03 for 4x1kg spool units changer, totals

	Item	Qty	Info	2026-02 sample link
	Heat set insert M3xD5xL4	68	Same as Voron BOM size	
	Flat Head Cap Screw M3x6	4		
	Flat Head Cap Screw M3x8	24		
	Socket Head Cap Screw M3x6	8		
	Socket Head Cap Screw M3x10	8		
	Socket Head Cap Screw M3x14	4		
	Socket Head Cap Screw M3x20	16		
	Socket Head Cap Screw M3x25	4		
	Socket Head Cap Screw M3x30	12		
	Self tapping screw M2x5	6		
	Self tapping screw M2x8 flat tip	36		
	Nut M12	4	M12x1.75 DIN934	
	Washer M12	4	M12x24x2.5 DIN125	
	Threaded rod M12	2	M12x1.75 DIN975, length depends on the changer and printer setups	
	BMG 17 teeth extruder gear	4	Extruder drive gear with grub screw for 5mm stepper shaft	
	Ball bearing MR63zz	8+1	Included 8 in 4x HGX-lite extruder part kits, 1 extra needed	
	Thumb screw assy with spring	4+4	Included 4 in 4x HGX-lite extruder part kits, 4 extras needed	
	Large extruder gear	8	Included in 4x HGX-lite extruder part kits	
	Needle bearing 3x5x6.5mm	8	Included in 4x HGX-lite extruder part kits	
	Bearing pin 3x16mm	8	Included in 4x HGX-lite extruder part kits	
	Bearing pin 3x12mm	12		
	Tension spring D6xL20xWD0.4	1	Tension spring for buffer slide (Total length with eyelets 6+8+6=20mm)	
	PTFE tube OD4mm ID2mm	1m		
	PTFE tube OD6mm ID4mm	3m	3 meter continuous lot	
	Tube collet 4mm	5		
	Tube collet 6mm	9		
	Cable ties, small	10	1.8mm tie width, 3.5mm eye width	
	Bicycle inner tube	≈0.3m	Optimal flat width ≈40-45mm to be stretched on a 32mm diameter roller, 8x35mm pieces	

	Item	Qty	Info	2026-02 sample link
	Stepper motor NEMA17 L=38mm	4	Optimal L=38(40)mm can be longer but affects wiring length and units sideways stacking fitment	
	Led SK6182 RGBW	4		
	Micro switch D2F	9		
	Micro switch D2F-L	2		
	JST PH2.0 3-m-pin housing	12	Switch and led inlet drive unit	
	JST PH2.0 4-m-pin housing	5	Motor inlet drive unit (4) and hub/buffer (1)	
	JST PH2.0 6-fm-pin housing	4	Stepper motor connector	
	JST PH2.0 3-fm-pin housing	12	Wirings on drive unit ends	
	JST PH2.0 4-fm-pin housing	5	Wirings on drive unit and hub/buffer ends	
	JST PH2.0 male pin			
	JST PH2.0 female pin			
	FEP wire UL1332 AWG26		OD max 1.1mm to fit wire bundles to channels in 3d printed parts at some tight places	
	Controller board*	1	For 4 spool units changer: a board with 4 stepper drivers, minimum 11 switch inputs and rgb	